PRACTICE EXAM 09 / EXAM SET 02

Domain 2: Anatomy and Physiology

8 scored questions	4 answer choices per question	Answer key + explanations included

Instructions

Choose the single best answer (A, B, C, or D) for each question. Mark your responses before checking the answer key and detailed explanations at the end of this document. These practice questions are not official NHA examination items.

QUESTIONS

Q01 | Body Structures and Organ Systems

> **SCENARIO** An older adult with a provider-diagnosed vestibular disorder is scheduled for balance rehabilitation. The patient asks how a problem in the ear can affect balance even when hearing seems unchanged. The medical assistant reinforces the provider's explanation.

Which explanation correctly identifies the involved structures and their normal function?

A. The cochlea detects head movement and sends balance signals to the brain.
B. The middle-ear bones detect body position and regulate leg muscle strength.
C. The external ear canal detects head rotation and controls eye movements.
D. The vestibular structures in the inner ear detect head movement and position.

Q02 | Pathophysiology and Disease Processes

> **SCENARIO** A patient with diagnosed gastroesophageal reflux disease asks why stomach contents sometimes reach the throat after a meal. The medical assistant uses the provider's teaching sheet to explain the underlying process.

Which change allows this backward movement to occur?

A. The upper esophageal sphincter remains closed during swallowing.
B. The lower esophageal sphincter relaxes inappropriately or closes inadequately.
C. The pyloric sphincter allows food to pass from the stomach into the duodenum.
D. The ileocecal valve allows intestinal contents to enter the large intestine.

Q03 | Body Structures and Organ Systems

SCENARIO A patient with provider-diagnosed emphysema is reviewing a lung diagram with the medical assistant. The provider has explained that destruction of thin-walled air sacs reduces the surface available for oxygen to move into the blood.

Which structures are the air sacs described in the explanation?

A. Bronchi
B. Alveoli
C. Tracheal rings
D. Pleural membranes

Q04 | Pathophysiology and Disease Processes

SCENARIO An older adult receiving long-term prescribed oral glucocorticoid therapy for a chronic condition has an order for dual-energy x-ray absorptiometry of the hip and spine. The patient says, "I do not have any bone pain, so I do not understand this test." The medical assistant reinforces the provider's explanation.

Which statement best explains the purpose of this ordered test?

A. It measures joint inflammation caused by cartilage irritation.
B. It measures muscle strength to identify the cause of falls.
C. It measures bone mineral density to help assess bone loss and fracture risk.
D. It measures blood calcium to determine whether a fracture has healed.

Q05 | Body Structures and Organ Systems

SCENARIO A patient is scheduled for a provider-performed Pap test. When reviewing the approved procedure information, the patient asks whether the sample comes from the ovaries. The medical assistant clarifies the anatomic source of the ordered sample.

Which structure is normally sampled for this test?

A. The cervix, the lower part of the uterus that opens into the vagina.
B. The endometrium, the tissue lining the inside of the uterine cavity.
C. The ovary, the organ that releases eggs and produces reproductive hormones.
D. The fallopian tube, the passage connecting the region of the ovary with the uterus.

Q06 | Pathophysiology and Disease Processes

SCENARIO A patient with type 2 diabetes and documented peripheral neuropathy reports a new blister on the sole of one foot but says, "It cannot be important because it does not hurt." The medical assistant reports the blister to the provider.

Which disease-related mechanism explains why the lack of pain does not establish that the injury is minor?

A. Reduced arterial circulation slows wound healing by limiting oxygen delivery to the foot.
B. Joint cartilage degeneration reduces cushioning between bones during weight-bearing.
C. Peripheral sensory nerve damage can reduce awareness of pain from a foot injury.
D. Elevated blood glucose can increase the risk of infection in an existing wound.

Q07 | Body Structures and Organ Systems

SCENARIO A healthy adult has been sweating during outdoor work and has taken in little water. During an educational discussion, the provider explains that antidiuretic hormone release increases as the body conserves water. The medical assistant reinforces this explanation.

Which kidney response normally follows an increase in antidiuretic hormone?

A. Less water is reabsorbed into the blood, producing a larger volume of dilute urine.
B. More water is reabsorbed into the blood, producing a smaller volume of concentrated urine.
C. More glucose is removed from the blood, producing a smaller volume of glucose-rich urine.
D. Less sodium is filtered from the blood, producing a larger volume of concentrated urine.

Q08 | Pathophysiology and Disease Processes

Which process most directly produces the joint inflammation characteristic of rheumatoid arthritis?

A. Uric acid crystals collect in the joint and trigger local inflammation.
B. Repeated mechanical loading primarily damages articular cartilage without an autoimmune attack.
C. Bacteria multiply in the joint space and produce a localized infection.
D. An autoimmune response causes inflammation of the synovial lining.

ANSWER KEY & EXPLANATIONS

Practice Exam 09 | Domain 2: Anatomy and Physiology | 8 scored questions

01 D	02 B	03 B	04 C	05 A	06 C
07 B	08 D				

DETAILED EXPLANATIONS

Q01 | Correct answer: D

The vestibular structures in the inner ear detect head movement and position.

The vestibular organs of the inner ear provide information about head movement and position that the brain uses to maintain balance. The cochlea primarily converts sound into nerve signals, while the middle-ear bones transmit sound vibrations. A vestibular disorder can therefore affect balance without necessarily causing a perceived hearing change.

Q02 | Correct answer: B

The lower esophageal sphincter relaxes inappropriately or closes inadequately.

The lower esophageal sphincter normally helps prevent stomach contents from moving back into the esophagus. Weakness or inappropriate relaxation can allow reflux, which may reach the throat. The pyloric sphincter controls passage toward the small intestine and does not form the barrier between the stomach and esophagus.

Q03 | Correct answer: B

Alveoli

Alveoli are small air sacs surrounded by capillaries, where oxygen and carbon dioxide move between air and blood. Loss of alveolar walls reduces the surface available for this exchange. Bronchi conduct air toward the gas-exchange areas but are not the thin-walled air sacs described.

Q04 | Correct answer: C

It measures bone mineral density to help assess bone loss and fracture risk.

Dual-energy x-ray absorptiometry measures bone mineral density and helps providers evaluate osteoporosis and fracture risk. Bone loss can occur without pain, and long-term glucocorticoid use can increase the risk. This test does not measure muscle strength, joint inflammation, or blood calcium.

Q05 | Correct answer: A

The cervix, the lower part of the uterus that opens into the vagina.

A Pap test samples cells from the cervix to look for abnormal cellular changes. The cervix is the lower portion of the uterus that opens into the vagina. The ovaries and endometrial lining are different structures and are not the routine sampling site for a Pap test.

Q06 | Correct answer: C

Peripheral sensory nerve damage can reduce awareness of pain from a foot injury.

Peripheral neuropathy can damage sensory nerves and reduce protective sensation in the feet. A patient may therefore have a blister or other injury without noticing pain, so the absence of pain does not establish that tissue injury is minor. Arterial disease may also affect a diabetic foot, but it is not the mechanism of sensory loss specified here.

Q07 | Correct answer: B

More water is reabsorbed into the blood, producing a smaller volume of concentrated urine.

Antidiuretic hormone, also called vasopressin, signals the kidneys to conserve water by increasing water reabsorption. With otherwise normal kidney function, this produces less urine that is more concentrated, helping maintain fluid balance. Increased loss of dilute urine would oppose water conservation and is the opposite response.

Q08 | Correct answer: D

An autoimmune response causes inflammation of the synovial lining.

Rheumatoid arthritis is an autoimmune disease in which immune activity causes inflammation of the synovial lining and can damage joint structures. Uric acid crystal deposition describes gout, while bacterial infection describes septic arthritis. Primarily degenerative cartilage damage is associated with osteoarthritis rather than the central autoimmune process in rheumatoid arthritis.

End of Practice Exam